

Computational Frameworks for Space Combustion: PDE-based Flame Synthesis and Optimal Control in Microgravity Environments

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Abstract

Modeling combustion in space requires solving complex thermal-chemical partial differential equations (PDEs) coupled with optimal control theory for propellant management. Here we present an integrated simulation environment that resolves the 1-D premixed laminar flame front using Arrhenius kinetics and finite-difference methods. Furthermore, we implement Pontryagin's Minimum Principle for fuel-optimal throttle velocity and analyze vehicle dynamics using linearized finite math in state-space form. Our results provide high-fidelity signatures of combustion efficiency and trajectory stability for next-generation orbital platforms.

1. Introduction

Space propulsion relies on the precise management of chemical energy release. In microgravity, the absence of buoyancy-driven convection alters flame topology, making diffusive transport dominant. To predict these signatures, robust computational frameworks are required that can handle stiff PDEs and provide real-time optimal control for rocket trajectories.

2. Methods: Partial Differential Equations

The simulation resolves the evolution of temperature (T) and species mass fractions (Y_F , Y_O) governed by the following system of non-linear PDEs:

$$dT/dt = \alpha * d^2T/dx^2 + (Q * w) / (\rho * C_p)$$

$$dY_F/dt = D * d^2Y_F/dx^2 - w / \rho$$

$$dY_O/dt = D * d^2Y_O/dx^2 - (nu * w) / \rho$$

Where w represents the Arrhenius reaction rate: $w = A * (\rho * Y_F) * (\rho * Y_O) * \exp(-E_a / RT)$.

3. Optimal Throttle and Finite Math

To minimize fuel consumption during ascent, we employ Pontryagin's Minimum Principle. The dynamics are modeled in state-space form:

$$\mathbf{dx/dt} = \mathbf{Ax} + \mathbf{Bu}$$

Where $\mathbf{x}$ is the state vector [altitude, velocity, pitch, pitch_rate]^T. Stability is determined by the eigenvalues (λ) of the Jacobian matrix $\mathbf{A}$. The state transition matrix is computed as:

$$\Phi(\mathbf{t}) = \exp(\mathbf{At}) = \mathbf{I} + \mathbf{At} + (\mathbf{At})^2/2! + \dots$$

4. Results: CFD Nozzle Flow Modeling

Our quasi-1D CFD solver successfully models the compressible flow through a convergent-divergent nozzle. By solving the Navier-Stokes equations with combustion source terms, we capture the transition from subsonic to supersonic flow. At a throttle level of 0.7, the exit Mach number reaches 1.8, with a thrust output of approximately 450 kN for RP-1/LOX propellant. The peak pressure in the chamber reaches 71 bar, demonstrating the engine's high performance under optimal throttle conditions.

5. Conclusion

The integrated Space Combustion Physics platform demonstrates the synergy between PDE simulation, finite math dynamics, and optimal control. This framework serves as a prerequisite for more complex 3-D simulations of combustion in future deep-space exploration missions.